EXPERIMENT 11: FUZZY INFERENCE SYSTEM

AIM

To implement a fuzzy inference system for temperature control and determine fan speed from a given temperature.

PROCEDURE

1. Define the temperature and fan-speed ranges with membership functions.

2. Fuzzify the input temperature of 28°C and apply the low, medium, and high rules.

3. Combine the output membership functions and use centroid defuzzification.

4. Display the crisp fan-speed result.

CODE

Python Program

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

temperature = np.arange(0, 41, 1)
fan_speed = np.arange(0, 11, 1)

# Membership functions for temperature
temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
temp_high = fuzz.trimf(temperature, [20, 40, 40])

# Membership functions for fan speed
fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

# Fuzzify input temperature
temperature_input = 28
low_level = fuzz.interp_membership(temperature, temp_low, temperature_input)
medium_level = fuzz.interp_membership(temperature, temp_medium, temperature_input)
high_level = fuzz.interp_membership(temperature, temp_high, temperature_input)

# Apply rules and aggregate output membership functions
aggregated = np.fmax(
    np.fmax(np.minimum(fan_low, low_level),
            np.minimum(fan_medium, medium_level)),
    np.minimum(fan_high, high_level)
)

# Defuzzification using centroid method
fan_output = fuzz.defuzz(fan_speed, aggregated, 'centroid')

print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")
```

OUTPUT

```
Temperature: 28°C
Fuzzified fan speed: 7.17
```

RESULT

Thus, a fuzzy inference system was implemented to map the input temperature to a crisp fan-speed output.